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Manipulative Therapy Effectiveness Following Acute Lateral Ankle Sprains

A Systematic Review

Erik A. Wikstrom, PhD, ATC; and Patrick O. McKeon, PhD, ATC

■ ABSTRACT

Manipulative therapies, such as high-velocity low-amplitude thrusts and joint mobilizations, are common therapeutic interventions for lateral ankle sprains and are purported to improve numerous outcome measures. However, there is limited evidence indicating that these techniques improve outcomes in this population. Therefore, a systematic review was conducted. Three studies were included, and effect sizes and 95% confidence intervals were calculated for pain, range of motion, and functional outcomes measures. The results indicate that a single session of manipulative therapy failed to improve the studied outcome measures. However, multiple sessions of manipulative therapy reduced pain, improved range of motion, and improved function in patients with acute lateral ankle sprains. To date, no investigation has directly compared different manipulative therapy techniques, thus the most effective technique remains unknown. The cumulative findings indicate that multiple sessions of manipulative therapy are an effective modality in treating the symptoms of acute lateral ankle sprains.

Lateral ankle sprains are painful, incapacitating,^{1,2} and among the most common musculoskeletal injuries.^{3,4} Further, decreased dorsiflexion

range of motion (ROM)^{5,6} and altered arthrokinematics of the talocrural joint⁷ have been reported following lateral ankle sprains, and these adaptations have been hypothesized to be contributing factors for the functional impairments associated with acute lateral ankle sprains.^{5,8,9} Most importantly, failure to properly treat acute lateral ankle sprains and resolve the above mentioned symptoms can result in serious long-term sequela, such as chronic ankle instability¹⁰ and posttraumatic ankle osteoarthritis.¹¹ The seriousness of these consequences demonstrates the importance of identifying therapeutic modalities that can effectively treat both the mechanical (ie, ROM and arthrokinematics) and functional (ie, gait and self-reported disability) consequences of acute lateral ankle sprains.

Manipulative therapy techniques are becoming more widely accepted in clinical practice as treatments for the negative sequelae of acute lateral ankle sprains. To date, both anecdotal evidence and narrative reviews have indicated that manipulative therapy techniques of the talocrural joint, including Maitland's mobilizations (oscillatory in nature),^{12,13} Mulligan's mobilizations with movement (sustained glide with active motion),¹⁴⁻¹⁶ and high-velocity low-amplitude (HVLA) thrusts,¹⁷⁻²⁰ may be effective treatments for acute lateral ankle sprains. These preliminary findings make intuitive sense because evidence indicates that the posterior glide of the talus during dorsiflexion is decreased following acute lateral ankle sprains⁷ and following prolonged immobilization of the ankle joint,²¹ and manipulative therapies associated with the talocrural joint are designed to reduce pain and improve ROM via the restoration of arthrokinematic motions (ie, roll, glide, spin).¹² Therefore, it would appear that manipulative

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TABLE
Characteristics of the Articles Included in the Systematic Review

ARTICLE	PEDro SCORE	STUDY DESIGN	INCLUSION CRITERIA	MANIPULATIVE THERAPY	PLACEBO	CONTROL	THERAPY TECHNIQUE	NO. OF SESSIONS	DURATION	FOLLOW-UP
Pellow & Brantingham ³²	4	Single-blind, comparative controlled pilot study	Grade 1 or 2 LAS	HVLA thrust (n=15)	Detuned ultrasound (n=15)	n/a	Single HVLA thrust	Maximum of 8	Maximum of 4 weeks	1 month
Collins et al ²⁴	5	Double-blind, randomized controlled cross-over trial	Grade 2 LAS	MWM (n=16)	"Laying of hands" (n=16)	Same position (n=16)	3 sets of 10 MWM repetition	1	n/a	None
Green et al ⁹	7	Randomized controlled trial	Acute ankle sprains	Oscillatory mobilizations & RICE (n=19)	n/a	RICE (n=19)	3 sets of oscillatory mobilizations lasting 60 seconds	Maximum of 6	Maximum of 14 days	None

Abbreviations: LAS, lateral ankle sprain; HVLA, high-velocity low-amplitude; MWM, mobilization with movement; RICE, Rest, Ice, Compression, Elevation.

therapies of the talocrural joint could be an effective treatment for acute lateral ankle sprains.

However, it remains unclear whether there are any substantial benefits derived from using manipulative therapy of the talocrural joint to treat acute lateral ankle sprains for several reasons. For example, a variety of outcome measures have been reported in the literature, and many investigations have used multiple manipulative therapy techniques simultaneously, making it difficult to interpret the individual effect of each type of manipulative therapy on acute lateral ankle sprains. Moreover, this has made it extremely difficult to reach a consensus regarding best practice. Therefore, the purpose of this systematic review was to answer the following clinical question: Does manipulative therapy of the talocrural joint improve outcome measures in patients with acute lateral ankle sprains? By systematically reviewing the existing literature and reporting the mutually exclusive effect sizes of manipulative therapies, as well as their confidence intervals, this review expands on previous work and provides further evidence regarding best practice.

LITERATURE REVIEW

Search Strategy

We searched PubMed, CINAHL, and SPORTDiscus entries between 1980 and 2010 using the terms *ankle sprain*, *lateral ankle sprain*, *manipulation*, *mobilization*, and *manual therapy*. In addition, the reference lists of retrieved articles were examined and a hand search was conducted. Our search identified 24 potential articles on the effects of manipulative therapy on the talocrural joint.^{9,13-15,17-20,22-37} Criteria for inclusion and exclusion were determined by our leading research question. For inclusion, the studies needed to meet 3 specific criteria: use of patients with acute lateral ankle sprains, the mutually exclusive use of a manipulative therapy technique of the talocrural joint (ie, use of a single technique), and the comparison of the manipulative therapy technique against a control and/or placebo group. For the purposes of this investigation, we operationally defined *acute* as an injury that occurred within 8 weeks of enrollment in the original investigation. We chose to investigate only studies that used a single manipulative technique on the talocrural joint because we were interested in determining whether any manipulative therapy technique was effective in this population.

Criteria for Study Selection

A summary of our search and data extraction procedure follows: both authors conducted an exhaustive search for pertinent studies; inclusion and exclusion decisions for each study in the systematic review were determined with unanimous consensus; and data were extracted and analyzed independently by each author. If differences were noted, a discussion occurred to reach a consensus.

Twenty-one articles were excluded from our analysis for the following reasons: 5 were review articles,^{13,17-20} 4 were case studies/series,^{14,15,25,36} 4 used multiple manipulative therapy techniques,^{23,27,30,37} 4 used uninjured or asymptomatic patients,^{26,28,31,34} 2 used patients with chronic ankle instability,^{29,35} and 2 failed to meet our operational definition of an acute ankle sprain.^{22,33} The Table provides detailed characteristics of the three included studies.^{9,24,32}

Quality of Included Investigations

For this systematic review, all included studies were evaluated using the PEDro scale,³⁸ a 10-point assessment tool used to evaluate the methodological quality of original research investigations. Consensus PEDro scores were agreed on after each author independently scored each of the included articles. A higher PEDro score indicates a higher quality study design. The methodological quality of each included investigation is provided in the Table.

Outcome Measures

The specific outcome measures varied across studies but all fell under 1 of 3 categories: pain, ROM, and function. Specific pain variables included self-reported pain and pain pressure thresholds. Range of motion variables included active and passive nonweight-bearing dorsiflexion, as well as weight bearing dorsiflexion ROM. Functional outcome measures included spatio-temporal gait parameters and self-reported function.

To determine the effect of manipulative therapies on the pain, ROM, and function variables following acute lateral ankle sprains, standardized effect sizes (Cohen *d*) and 95% confidence intervals (CIs) were calculated for changes from pretreatment to posttreatment in each outcome measure reported, as described in the original investigations. A positive effect size indicates an improvement in the outcome measure following the treatment session. On the contrary, a negative effect

size indicates that the outcome measures worsened following treatment. The strength of the effect sizes was interpreted using the guidelines described by Cohen, with values less than 0.4 interpreted as weak, between 0.41 and 0.7 as moderate, and greater than 0.7 as strong.³⁹

The quality of evidence used to answer the clinical question was assessed using the Strength of Recommendation Taxonomy (SORT).⁴⁰ The SORT level of recommendations range from 1 to 3, where 1 indicates good quality patient-oriented evidence, 2 indicates limited quality patient-oriented evidence, and 3 indicates nonpatient-oriented evidence. The SORT strength of recommendations range from A to C. A strength of recommendation of A indicates the recommendation is based on consistent and good quality patient-oriented evidence, B indicates that the recommendation is based on inconsistent or limited quality patient-oriented evidence, and C indicates that the recommendation is based on evidence other than patient-oriented evidence.^{8,41}

FINDINGS

Does manipulative therapy of the talocrural joint improve outcome measures in patients with acute lateral ankle sprains?

Pain

Two of the three included investigations, with a mean PEDro score of 4.5, examined pain pressure scores (ie, point tenderness)^{24,32}; one of these investigations, with a PEDro score of 4, examined changes in self-reported pain.³² The point estimates for pain pressure effect sizes ranged from -0.14 to 1.55, with positive scores indicating that the treatment elevated the pain pressure threshold (ie, the patient was less point tender). A single manipulative therapy treatment resulted in small positive effects on point tenderness but so did the corresponding placebo and control treatments (Figure 1). Further, the CIs of the experimental, placebo, and control groups all crossed zero, indicating that the true effect of a single session of manipulative therapy is unclear.

Manipulative therapy treatments delivered over several treatment sessions (maximum of 8 in 4 weeks), as would be done in a clinical practice, resulted in large positive effect sizes for both point tenderness (Figure 1) and self-reported pain (Figure 2). Although the placebo group also saw large positive effect sizes,

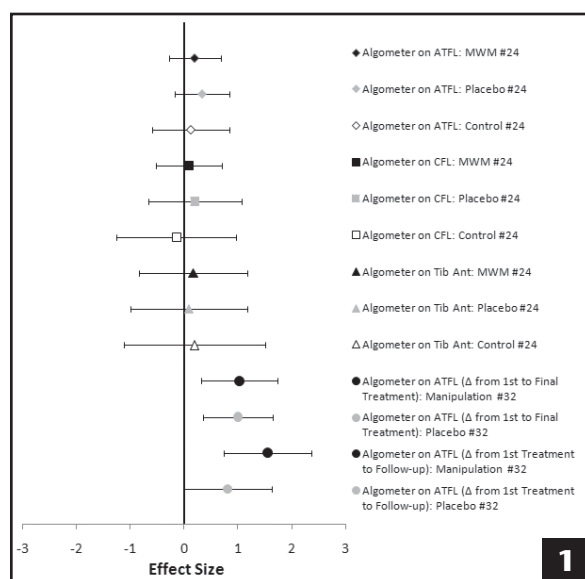


Figure 1. Forest plot of pain pressure threshold outcome measures reported by Pellow and Brantingham³² and Collins et al.²⁴ Abbreviations: ATFL, anterior talofibular ligament; CFL, calcaneofibular ligament; Tib Ant, tibialis anterior.

the manipulative therapy group consistently reported larger effect sizes, suggesting that the treatment is effective. In addition, the CIs for point tenderness and pain quality (McGill Pain Questionnaire) following the final manipulative therapy treatment and at the 1-month follow up did not cross zero. These findings indicate that a positive effect of pain outcome measures occurred following multiple sessions involving HVLA thrust treatments in patients with acute lateral ankle sprains.

Based on this evidence, it appears that a single manipulative therapy treatment is not effective at altering pain outcomes in individuals with acute lateral ankle sprains. However, multiple HVLA thrust treatments appear to be effective because pain pressure scores and subjective pain scores improved in individuals with acute lateral ankle sprains. For this finding within the answer to our clinical question, the SORT level of evidence was 2 because these were low-quality prospective studies. The strength of recommendation for the incorporation of this evidence into clinical practice was B based on the low-quality patient-oriented evidence and the limited number of data points with consistent results across studies.

Range of Motion

All 3 investigations^{9,24,32} quantified changes in dorsiflexion ROM and had a mean PEDro score of 5.33.

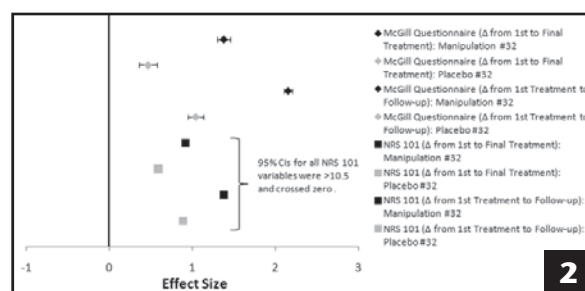


Figure 2. Forest plot of self-reported pain outcome measures reported by Pellow and Brantingham.³² Abbreviations: NRS, numerical rating scale; CI, confidence interval.

Active ROM³² point estimates ranged from 0.32 to 1.97, but all CIs crossed zero; therefore, it is unclear whether multiple HVLA thrust treatments improve active dorsiflexion ROM (Figure 3), but the consistently larger point estimates for the experimental group, relative to the placebo group, suggest the treatment may be beneficial. Passive ROM⁹ point estimates ranged from 0.41 to 9.55 and only a single CI (control group after a single treatment session) crossed zero (Figure 4), suggesting that oscillatory mobilizations can improve passive dorsiflexion ROM. In addition, there appears to be a cumulative effect of multiple oscillatory mobilization treatments because the point estimate for a single treatment was smaller than all of the point estimates for multiple treatments. Point estimates for weight bearing ROM (Figure 5), as tested with a lunge test,²⁴ range from -0.06 to 0.27 and all CIs crossed zero. Thus, the effect of a single mobilization with movement (MWM) treatment session, if any, is unclear.

Based on this evidence, it appears that a single manipulative therapy treatment is not effective at improving weight bearing ROM. However, multiple manipulative therapy treatments do appear to improve both active and passive ROM, but the evidence regarding active ROM is still controversial because the CIs crossed zero. For this finding within the answer to our clinical question, the SORT level of evidence was 2, with a strength of recommendation for the incorporation of this evidence into clinical practice of B because of the low quality patient-oriented evidence and limited number of data points with consistent results across studies.

Function

Two of the three included studies, with a mean PEDro score of 5.5, quantified the effects of manipulative thera-

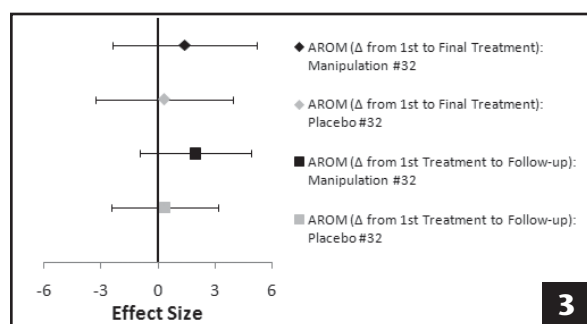


Figure 3. Forest plot of active range of motion outcome measures reported by Pellow and Brantingham.³² Abbreviation: AROM, active range of motion.

py on function.^{9,32} The point estimates for improvements in spatiotemporal gait parameters⁹ ranged from -0.05 to 2.82 , with the majority of point estimates representing moderate to large positive effects (Figure 6). Furthermore, CIs crossed zero only following the first treatment, which indicates that the true effect of a single oscillatory mobilization is unclear and that multiple oscillatory mobilization treatments can improve gait parameters in individuals with an acute lateral ankle sprain.

Point estimates for self-reported function³² ranged from 0.76 to 2.37 (Figure 7). All point estimates represent large positive effects but all CIs crossed zero, indicating that the true effect of multiple HVLA thrust treatments on self-reported function is unclear. However, a trend that HVLA thrust treatments are more effective than a placebo treatment was present based on the larger effect sizes. Based on this evidence, it appears that multiple manipulative therapy treatments are effective at improving functional measures. For this finding within the answer to our clinical question, the SORT level of evidence was 2, with a grade of recommendation of B based on the low-quality patient-oriented evidence and the limited number of data points with fairly consistent results across studies.

DISCUSSION

Does manipulative therapy of the talocrural joint improve outcome measures in patients with acute lateral ankle sprains? The most important finding of this systematic review is that multiple treatment sessions of manipulative therapy of the talocrural joint are needed to reduce pain, increase range of motion, and improve functional outcome measures. A single treatment session has no effect on pain, range of motion, and functional outcomes. To date, much of the information regarding the effectiveness of manipulative therapies on

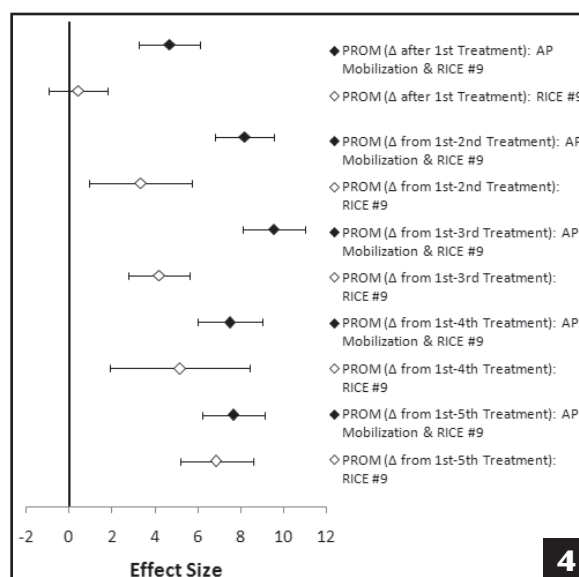


Figure 4. Forest plot of passive range of motion outcome measures reported by Green et al.⁹ Abbreviations: PROM, passive range of motion; AP, anterior to posterior; RICE, Rest, Ice, Compression, Elevation.

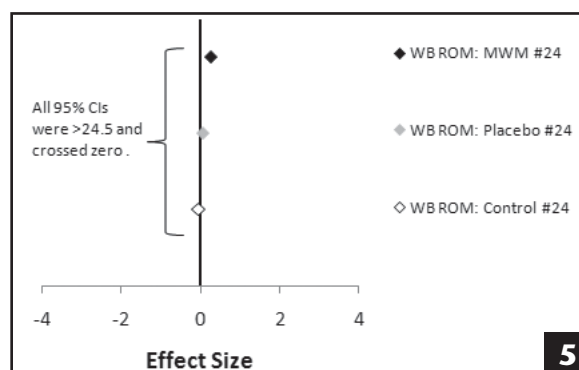


Figure 5. Forest plot of weight-bearing range of motion outcome measures reported by Collins et al.²⁴ Abbreviations: WB ROM, weight bearing range of motion; MWM, mobilization with movement; CI, confidence interval.

the treatment of acute ankle sprains is anecdotal and derived from case studies.^{14,15,25,36} These case reports typically indicate that significant and “immediate” improvements are seen after a manipulative therapy treatment session, which is contrary to the current findings.

Pain

Based on the evidence, it appears that a single MWM treatment does not improve pain pressure thresholds.²⁴ However, 2 strong trends of improvement were seen in pain pressure scores and self-report levels of pain following multiple HVLA thrust treatments.³² Indeed,

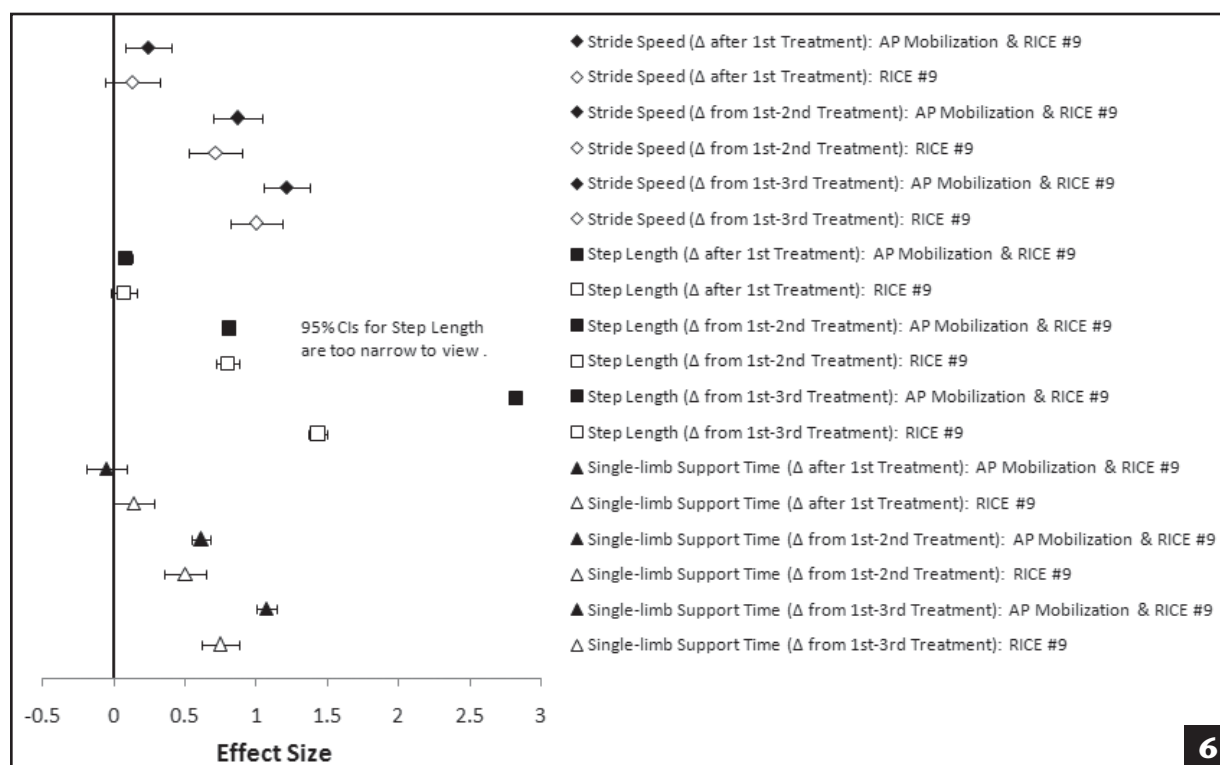


Figure 6. Forest plot of spatiotemporal gait parameters reported by Green et al.⁹ Abbreviations: AP, anterior to posterior; RICE, Rest, Ice, Compression, Elevation; CI, confidence interval.

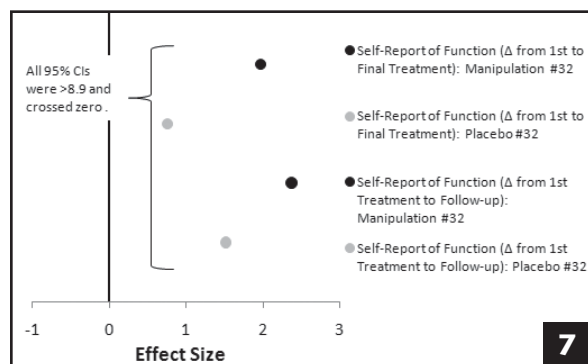


Figure 7. Forest plot of self-reported function outcome measures reported by Pellow and Brantingham.³² Abbreviation: CI, confidence interval.

the effect sizes of the manipulative therapy group were consistently larger than those of the placebo group, and the CIs of the pain pressure data points did not cross zero for the HVLA thrust treatment group. In addition, the large positive effects associated with the placebo group may be due to the natural recovery process over the 4-week study duration.

Previously, a single treatment session, which involved multiple osteopathic and manipulative techniques, immediately reduced self-reported pain rela-

tive to a control group.²⁷ Although contrary to our findings, it is possible that the multiple treatment techniques within the same session resulted in a cumulative effect similar to the effects seen after multiple treatment sessions involving a single manipulative therapy technique. Similarly, investigations^{23,37} involving multiple manipulative techniques delivered over several treatment sessions have reported improvements in pain outcome measures.

For example, multiple HVLA thrust treatments and oscillatory techniques used in conjunction with therapeutic exercise over a maximum of 3 treatment sessions on 85 acute lateral ankle sprains³⁷ resulted in decreased self-reported pain for patients who subjectively deemed the treatments successful (pretreatment: 4.0 ± 1.9 , posttreatment: 1.2 ± 0.9 , effect size: 1.47) and those that did not deem the treatment successful (pretreatment: 3.9 ± 2.3 , posttreatment: 2.0 ± 1.5 , effect size: 0.83). In addition, 6 sessions of multiple manipulation techniques in conjunction with cryotherapy delivered over a 2-week period reduced pain as effectively as a 7-day course of the nonsteroidal anti-inflammatory drug piroxicam and cryotherapy. However, neither treatment protocol

improved pain pressure thresholds (ie., point tenderness).²³

Thus, it appears that multiple manipulative therapy treatment sessions for the talocrural joint should be used to improve outcome measures associated with pain in patients with acute lateral ankle sprains. However, based on the limited evidence, the current investigation was unable to ascertain which manipulative therapy technique (oscillatory mobilizations, MWM, HVLA thrust) is most effective at improving pain outcomes because no investigation has directly compared treatment techniques against each other. Future research should determine the exact number of treatments and dosage within each treatment session needed to reduce pain.

Range of Motion

The current results suggest that both active and passive ROM are improved following the delivery of multiple oscillatory⁹ and HVLA thrust³² treatment sessions. However, weight bearing ROM did not improve following a single session of MWM.²⁴ Previously, the delivery of a variety of manipulative therapy techniques over the course of a 2-week intervention resulted in improved nonweight-bearing ROM.²³ However, a single treatment session, which included multiple manipulative techniques,²⁷ failed to improve ROM in patients with acute lateral ankle sprains. Similarly, a single HVLA thrust failed to improve ROM in patients who had experienced a lateral ankle sprains at least 6 months prior to testing²² and in asymptomatic controls.^{28,31} These findings suggest that ROM improvements will occur only in patients with acute lateral ankle sprains as a result of the cumulative effect of multiple manipulative therapy treatments.

Multiple treatment sessions of oscillatory mobilizations of the talocrural joint and HVLA thrust treatments improved ROM, but is one technique more effective? High-velocity low-amplitude thrust treatments resulted in large positive effect sizes, but the CIs crossed zero. Oscillatory mobilizations also caused large positive effect sizes, but the corresponding CIs did not cross zero. Thus, both techniques have the potential to be beneficial, but the benefits of HVLA thrust treatments remain unclear. Significant improvements in dorsiflexion ROM have also been reported after just a single treatment session of anterior-to-posterior talocrural mobilizations (oscil-

latory) in asymptomatic patients^{26,34} and in patients who underwent a prolonged period of ankle immobilization for a variety of pathological conditions.²¹ The combined data suggest that oscillatory mobilizations may have a more profound effect on ROM than do other manipulative therapy techniques, even after just a single treatment. However, this is speculative, and well designed randomized controlled trials are needed to systematically compare the effectiveness of oscillatory mobilizations and HVLA thrust treatments at improving dorsiflexion ROM in patients with acute lateral ankle sprains.

Function

Functional outcome measures also improved following multiple manipulative therapy treatments, specifically oscillatory mobilizations⁹ and HVLA thrusts.³² These findings are consistent with previously published reports. For example, 3 treatment sessions consisting of multiple HVLA thrust and oscillatory mobilization techniques significantly improved self-reported function in patients with acute lateral ankle sprains who subjectively felt the treatments were successful and those who felt the treatments were unsuccessful.³⁷ Indeed, these improvements were seen on 3 separate scales, including the Foot and Ankle Ability Measure Activities of Daily Living scale (successful Δ : +23.7, unsuccessful Δ : +12.6), Foot and Ankle Ability Measure Sport scale (successful Δ : +28.4, unsuccessful Δ : +11.5), and the Lower Extremity Functional Scale (successful Δ : +0.7, unsuccessful Δ : +1.9).

Six treatment sessions of multiple manipulative techniques also resulted in improved self-reported function following the final treatment.²³ There is also evidence suggesting that a single treatment session consisting of 2 manipulative therapy techniques can lead to an immediate redistribution of foot loading patterns during static stance relative to a placebo laying of hands procedure.³⁰ It is possible that the use of multiple treatment techniques was the cause of improved function after just a single treatment session, but this is purely speculative.

The current evidence indicates that HVLA thrust treatments and oscillatory mobilizations can improve function; however, the evidence suggests that oscillatory mobilizations may be superior because the associated CIs do not cross zero. Again, it is important to note that none of the included investigations directly

compared the effectiveness of different manipulative therapy techniques. Therefore, a definitive consensus regarding which techniques are most effective at improving functional outcomes will remain unknown until different techniques are directly compared. In addition, future research is needed to determine whether and which manipulative therapies can reduce or prevent recurrent injury following acute lateral ankle sprains.

CONCLUSION

Based on the level of evidence and strength of recommendation, for the answer to this clinical question to be incorporated into clinical practice it appears that multiple treatment sessions of manipulative therapy of the talocrural joint may have clinical benefit in the treatment outcomes for those who experience acute lateral ankle sprains. Specially, HVLA thrust treatments improved pain and demonstrated a strong trend for the improvement of ROM and function. In addition, oscillatory joint mobilizations improved ROM and function, but pain was not investigated. No investigation has quantified the effects of multiple MWM treatments on any outcome measure, but a single treatment session failed to improve pain or ROM outcomes. ■

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